

# Lecture 3: Elasticities & Taxes — the Supply & Demand model II (Ch. 2 recap · §6.4)

*Applied Microeconomics — Master Notes · by Jakob V. Stangel*

**Required reading:** Friberg, §6.4 Elasticities (own-price, cross-price, income; supply elasticity) — building on the Chapter 2 supply-and-demand model (see Lecture 2). *The lecture deck also teaches taxes on goods & services — that part draws on the textbook's §7.5 (unit tax, tax incidence), covered here because the deck foregrounds it.*

**Theme of the lecture** (Butera — from the deck, titled "*Supply and Demand model*"). Chapter 2 taught us to **move the curves around**; this lecture adds **precision**. The *"shape"* of a demand or supply curve — its *elasticity* — decides how much a shock (or a tax) shows up as a change in price versus a change in quantity. The deck's three goals: (1) understand what demand & supply elasticities mean, (2) derive them from demand/supply curves, and (3) use them to analyse the **effects of taxes**.

★ **The heart of the lecture:** elasticity = % responsiveness — "by what % does quantity change when price rises 1%." It's a **unit-free** number that summarises a curve's **shape**, so it's comparable across markets and time. The **shape decides how a shock splits into price vs quantity**: with **inelastic** demand a shock/tax hits **price**; with **elastic** demand it hits **quantity**. And a **tax** drives a **wedge** between the price buyers pay and sellers receive — **who bears it depends on the relative elasticities**, and (in perfect competition) **it doesn't matter who is legally taxed**.

**The red thread.** For any market ask: *how responsive is quantity to price (elastic or inelastic)?* — because that single question answers **how a supply shock moves the market**, **how a firm should price**, and **who really pays a tax**. Elasticity is the "shape" number that ties all three together.

## Key concepts / "modes" to use

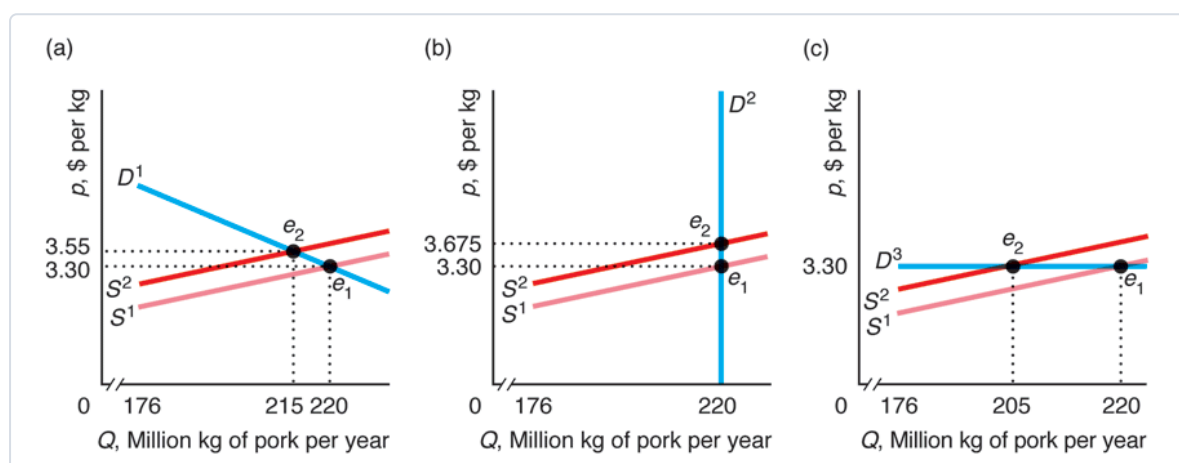
- **Elasticity** — the % change in one variable ÷ % change in another (a unit-free number).
- **Price elasticity of demand ( $E_D$ )** — % change in quantity demanded ÷ % change in own price (negative).
- **Elastic / inelastic / unit-elastic** —  $|E| > 1$  /  $|E| < 1$  /  $|E| = 1$ .
- **Point-elasticity trick** —  $E_D = (dQ/dp) \cdot (p/Q)$  ; on a linear curve elasticity **changes along the curve** (constant-elastic curves don't).
- **Cross-price elasticity** — substitutes (+) vs complements (-).
- **Income elasticity** — normal (+) vs inferior (-); **Engel curves / Engel's law**.
- **Supply elasticity** — % change in quantity supplied ÷ % change in price (positive).
- **Tax wedge & tax incidence** —  $p_b = p_s + t$  ; who bears the burden depends on **relative elasticities**; legal  $\neq$  economic incidence; unit vs ad-valorem tax; **deadweight loss**.

## Section-by-section main points

## Why do we care about elasticity? (deck)

**Core:** the **shape** of demand/supply curves determines **how much a shift in one moves the equilibrium** — and "shape" is summarised by **elasticity**. A steep (inelastic) curve and a flat (elastic) curve respond to the *same* shock very differently.

- **The motivating questions (deck):** Do "sin taxes" (cigarettes, sugar) hurt the poor? Why do firms **raise prices** even when it looks crazy (Netflix, Amazon Prime)? To what extent are **taxes on producers "passed on"** to consumers? Every answer runs through elasticity.
- **The visual intuition:** how far a **supply shift** moves price vs quantity depends entirely on the **slope/elasticity of demand** — see the figure below (perfectly inelastic demand → only price moves; perfectly elastic demand → only quantity moves).



**How to read it (why elasticity matters).** All three panels apply the **same supply shift**  $S^1 \rightarrow S^2$ . The **only** thing that differs is the **demand curve's shape**:

- **(a) Ordinary downward-sloping demand:** the shock splits into **both** a higher price *and* a lower quantity.
- **(b) Perfectly inelastic demand (vertical):** quantity can't respond, so the shock lands **entirely on price** ( $3.30 \rightarrow 3.675$ ). *This is the "necessity with no substitutes" case.*
- **(c) Perfectly elastic demand (horizontal):** price can't move, so the shock lands **entirely on quantity** ( $220 \rightarrow 205$ ).
- **Take-away:** the **more inelastic** the demand, the more a shock (or tax) shows up as **price**; the **more elastic**, the more it shows up as **quantity**. *This one picture is the whole reason we measure elasticity.*

## ★ Price elasticity of demand — definition

**Core:** the **price elasticity of demand** measures how sensitive quantity demanded is to the good's own price — the **% change in quantity** ÷ **% change in price**. It is a **pure (unit-free) number** and is **negative** (demand slopes down: price up → quantity down).

PRICE ELASTICITY OF DEMAND

$$E_D = ( \Delta Q / Q \times 100 ) \div ( \Delta P / P \times 100 ) = \% \Delta Q \div \% \Delta P$$

*Reads:* how responsive quantity demanded is to a price change — "a 1% rise in price  $p$  leads to an  $E_D$  % change in quantity  $q$ ." —  $\Delta Q/Q \times 100$  = the percentage change in quantity demanded ·  $\Delta p/p \times 100$  = the percentage change in price ·  $E_D$  = the ratio (a unit-free number, negative for a normal good). The  $\times 100$ s cancel, so it's just % $\Delta Q$  over % $\Delta p$ .

- **Why unit-free matters (the point of the whole concept):** raw slopes aren't comparable — "1,000 tons less rice" means something different in **Vietnam** vs **Bhutan**, and depends on the currency. A percentage measure is **comparable across markets, currencies and time**.
- **Why negative:** demand curves slope **down**, so  $\Delta Q$  and  $\Delta p$  have opposite signs → the ratio is negative. (When people say a "higher" elasticity they usually mean a bigger **absolute value**  $|E_D|$ .)
- **Worked example:** a café raises its coffee price **10%** and sells **8% fewer** cups →  $E_D = -8\% \div +10\% = -0.8$ . Since  $|-0.8| < 1$ , demand is **inelastic** — revenue actually *rises* (price up 10%, quantity down only 8%). (Full step-by-step computations are in the next section.)

## Computing elasticity — and the point-elasticity trick

*Core:* plug the two percentage changes into the formula. **On a linear demand curve the elasticity is not constant** — it depends on *where* you evaluate it, via  $E_D = (dQ/dp) \cdot (p/Q)$ .

- **Simple example (textbook Table 6.2):** quantity falls  $6 \rightarrow 4$  while price rises  $1 \rightarrow 2$ . Then

WORKED — SIMPLE EXAMPLE

$$E_D = ((4-6)/6) \div ((2-1)/1) = (-2/6) \div (1) = -1/3$$

*Reads:* a demand that is **inelastic** —  $(4-6)/6 = -0.33$  = the % fall in quantity (−33%) ·  $(2-1)/1 = 1$  = the % rise in price (+100%) ·  $-1/3$  = quantity falls **only 1/3 as fast** as price rises → **inelastic** demand.

- **A policy-style example (soda, Allcott et al. 2019):** own-price elasticity of sugary soda  $\approx -1.37$ . If price rises 5%, quantity changes by  $E_D \times 5\% = -1.37 \times 5 = -6.85\%$  (a ~7% fall). To cut demand 15%, price must rise  $-15 \div -1.37 \approx 11\%$ . → *this is exactly the sum a policymaker designing a soda tax needs.*
- **The point-elasticity trick (linear demand):** rewrite the formula as

POINT ELASTICITY

$$E_D = (\Delta Q/\Delta p) \cdot (p/Q) = (dQ/dp) \cdot (p/Q)$$

*Reads:* elasticity = **slope of the demand function** × the **price-to-quantity ratio at your chosen point** —  $dQ/dp$  = the **slope** (how many units  $Q$  changes per €1 of price; constant for a straight line) ·  $p/Q$  = the **price-to-quantity ratio** where you evaluate it (this is what varies along the curve).

- **Worked ( $Q = 10 - 2p$ ):** here  $dQ/dp = -2$  (always). But the **elasticity varies**:
  - at  $p = 2.5$  (so  $Q = 5$ ):  $E_D = -2 \times 2.5/5 = -1$  (**unit elastic**).
  - at  $p = 4$  (so  $Q = 2$ ):  $E_D = -2 \times (p/Q) = -2 \times 4/2 = -4$  → **elastic**. ( $\triangle$  The textbook misprints this as  $-2 \times 2/2 = -2$  — it accidentally puts  $Q=2$  in the numerator instead of  $p=4$ ; the correct  $p/Q$  is  $4/2 = 2$ , so  $E_D = -4$ .)

- **The pattern: demand is more elastic at higher prices** — at  $p=2.5$  it's unit-elastic ( $-1$ ), at  $p=4$  it's elastic ( $-4$ ). A fixed unit change is a bigger *percentage* change when  $Q$  is small and  $p$  is large.
- **Constant-elasticity demand (contrast):** the form  $Q = a \cdot p^\epsilon$  has the **same** elasticity  $\epsilon$  at *every* price (used in empirical estimation via  $\ln Q = \ln a + \epsilon \cdot \ln p + \text{error}$ ). So: **linear** → **elasticity changes along the curve**; **constant-elastic** → **it doesn't**.
- **Worked example:** with  $Q = 100 \cdot p^{-1.2}$ , the elasticity is  $-1.2$  whether you evaluate at  $p = €1$  ( $Q = 100$ ),  $p = €5$  ( $Q \approx 15$ ), or  $p = €50$  — it never changes. (Contrast the linear  $Q = 10 - 2p$  above, where it swung from  $-1$  to  $-4$ .)

## Elastic, inelastic, unit-elastic — and why it matters

**Core:** we **name** the ranges of elasticity, and each has a sharp economic meaning for **how supply shocks split into price vs quantity**.

| Name                | Level (of $E_D$ )            | Meaning                                         | Curve looks...          |
|---------------------|------------------------------|-------------------------------------------------|-------------------------|
| <b>Inelastic</b>    | $-1 < E_D < 0$ ( $ E  < 1$ ) | $\% \Delta Q$ <b>smaller</b> than $\% \Delta p$ | relatively <b>steep</b> |
| <b>Unit elastic</b> | $E_D = -1$                   | $\% \Delta Q$ <b>equals</b> $\% \Delta p$       | —                       |
| <b>Elastic</b>      | $E_D < -1$ ( $ E  > 1$ )     | $\% \Delta Q$ <b>bigger</b> than $\% \Delta p$  | relatively <b>flat</b>  |

- **Why it matters (the punchline):** with **inelastic** demand, **supply shocks translate into big price changes** (little quantity response) — critical for e.g. **cocoa** growers in Ghana/Côte d'Ivoire (world cocoa elasticity  $\approx -0.19$  to  $-0.96$ ; policy point estimate  $-0.34$ ). With **elastic** demand, shocks translate into **quantity** changes.
- **What makes demand elastic** — **substitutes**. The single biggest driver is **availability of close substitutes**: pharmaceuticals with few substitutes are **inelastic** (people pay whatever for a needed medicine); goods with many substitutes are **elastic**.
- **Aggregation level matters:** a *specific brand* is more **elastic** than the *category* (more substitutes for one brand). US beer: **product-level** median elasticity  $\approx -4.74$ , but beer **overall**  $\approx -0.60$ .

## Cross-price elasticity — substitutes vs complements

**Core:** the **cross-price elasticity** measures how demand for good **A** responds to a change in the price of a *different* good **B**. Its **sign** tells you the *relationship* between the two goods; its **size** tells you *how strong* that relationship is.

### CROSS-PRICE ELASTICITY

$$E_{cross} = \% \Delta Q_A \div \% \Delta p_B \rightarrow > 0 = \text{substitutes} \cdot < 0 = \text{complements} \cdot \approx 0 = \text{unrelated}$$

**Reads:** how the quantity of **A** responds to a price change in **B** —  $\% \Delta Q_A$  = percentage change in demand for good A ·  $\% \Delta p_B$  = percentage change in the price of good B ·  $E_{cross}$  = the ratio (can be + or −, unlike own-price which is always −).

- **The sign = the relationship:**
  - **Positive → substitutes.** B gets dearer → buyers switch to A → A's demand **rises** (butter ↑ → margarine demand ↑; Heineken ↑ → Carlsberg ↑).
  - **Negative → complements.** B gets dearer → buyers use less B *and* less of its partner A → A's demand **falls** (petrol ↑ → big-car demand ↓; smartphones ↑ → phone-case demand ↓).
  - **≈ 0 → unrelated goods** (the price of bread barely touches demand for haircuts).
- **The size = the strength:** the larger the **absolute value**  $|E_{cross}|$ , the **closer** the substitutes (or the stronger the complementarity). Two brands of the *same* soda have a **big** positive cross-price elasticity; soda and bottled water a **small** one; soda and umbrellas ≈ 0.
- **Worked example: Danish spirits get 10% dearer** → Swedish demand for spirits changes by  $E_{cross} \times 10\% = +0.3 \times 10 = +3\%$  (Swedes buy **more at home** instead of driving to Denmark) → **positive → substitutes** (Asplund 2007). *Flip it: if smartphones get 10% dearer and phone-case demand falls 6%,  $E_{cross} = -6 \div 10 = -0.6$  → negative → complements.*
- ★ **Its killer application — defining markets & judging mergers.** This is *the* reason cross-price elasticity matters in practice. Competition authorities use it to decide **whether two products are in the same market**: a **high positive** cross-price elasticity means A and B are **close substitutes**, so a firm selling both has real market power (a merger between them removes a competitor). This is the logic of the **SSNIP / "hypothetical monopolist" test** — *would a Small but Significant Non-transitory Increase in Price (say 5%) be profitable, or would enough buyers flee to substitutes to make it a loss?* If they flee, the substitute is in the same market.
- **Asymmetry (watch out):** cross-price elasticity is **not symmetric** — A's response to B's price ≠ B's response to A's price. A cheap **complement** (phone case) reacts strongly to the price of the expensive main good (phone), but the phone barely reacts to the case's price.
- **Real numbers (Asplund et al. 2007):** Swedish **spirits +0.3, beer +0.5, wine +0.2** with respect to their **Danish** prices — all **positive**, i.e. all **substitutes** for cross-border shopping (beer the closest). (*Deck aside: cigarettes & alcohol ≈ -1 — strong complements, Krauss et al. 2014.*)

## Income elasticity — normal vs inferior goods, Engel curves

**Core:** the **income elasticity** measures how demand responds to **income** — its **sign** distinguishes **normal** from **inferior** goods.

### INCOME ELASTICITY

$$E_{income} = \% \Delta Q \div \% \Delta I \rightarrow > 0 = \text{normal good} \cdot < 0 = \text{inferior good}$$

**Reads:** how quantity demanded responds to an income change —  $\% \Delta Q$  = percentage change in quantity ·  $\% \Delta I$  = percentage change in income · **sign: positive → normal** (demand rises with income), **negative → inferior** (demand falls as income rises — e.g. instant noodles, store-brand goods).

- **Worked example:** a household's **income rises 10%** and its spirits purchases rise 14% →  $E_{income} = +14 \div +10 = +1.4$  → **normal and income-elastic** (a "luxury,"  $> 1$ ) (Asplund 2007). *Flip it: income +10% but instant-noodle demand -5% →  $E_{income} = -5 \div 10 = -0.5$  → inferior.*

- **Nuance:** a good can be **normal at low incomes and inferior at higher** ones (hostels for a student: more nights as income rises, then switch to hotels). Tastes vary by person.
- **Uses:** long-term **projections** (healthcare, appliances in low-income countries) and predicting the **business cycle** (pharma is income-insensitive → "defensive stock").
- **Engel curves / Engel's law** (Ernst Engel, 1800s Prussian statistician): plot demand (vertical) against income (horizontal) — **positive slope for normal goods, negative for inferior**. **Engel's law:** the **share of income spent on food falls as income rises** → the income elasticity of food is **less than 1**.

## Supply elasticity — same principle

**Core:** the **price elasticity of supply** measures how sensitive quantity *supplied* is to price — same formula, but **positive** (supply slopes up).

### PRICE ELASTICITY OF SUPPLY

$$E_S = \% \Delta Q_S \div \% \Delta p \quad (\text{positive})$$

*Reads:* how responsive quantity supplied is to a price change —  $\% \Delta Q_S$  = percentage change in quantity supplied ·  $\% \Delta p$  = percentage change in price ·  $E_S$  = the ratio (**positive**: higher price → more supplied). A **steep** supply curve is **inelastic** (hard to expand output quickly); a **flat** one is **elastic**.

- **Worked example:** the market price of new flats rises **20%**, but the quantity of new flats supplied rises only **4%** this year →  $E_S = +4 \div +20 = +0.2$  → **inelastic** supply (you can't build quickly). Give it several years and builders respond more → supply becomes **more elastic**.

## Elasticity over time — short run vs long run

**Core:** demand (and supply) are usually **more elastic in the long run** than the short run, because people have **more time to adjust / find substitutes**.

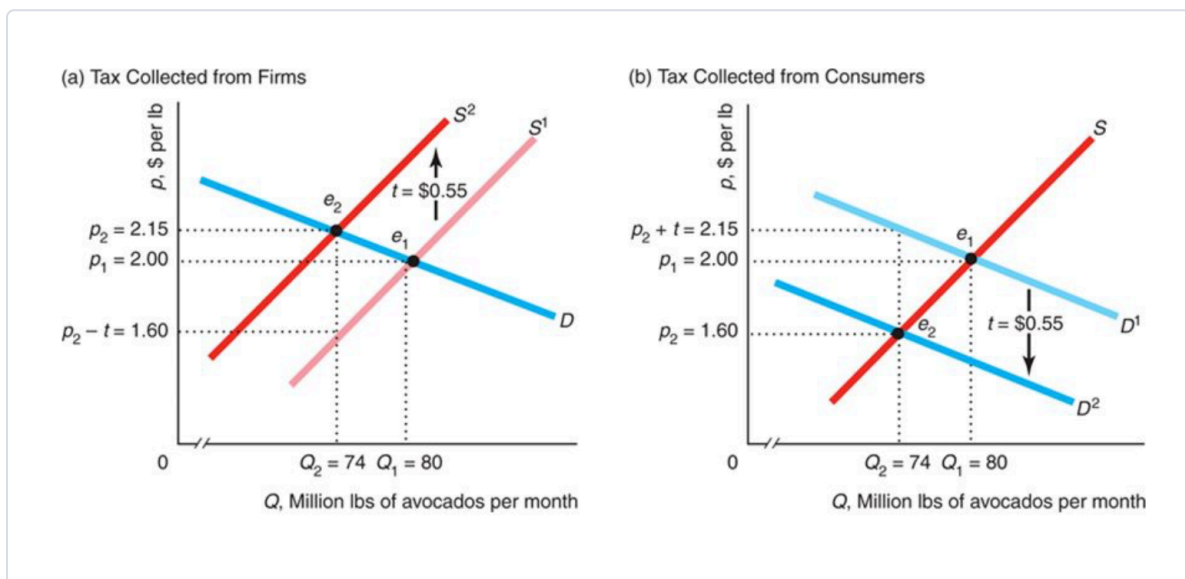
- **Electricity** (Labandeira et al. 2017):  $\approx -0.24$  short-run vs  $-0.6$  long-run — over time consumers buy more efficient equipment, change habits.
- **The deck's cases:** a drop in **gas** prices does little in the short run but more in the long run; likewise **computers**. *Exam framing:* "in the short run demand is inelastic (people are locked in); given time they substitute, so long-run demand is more elastic."

## ★ Taxes on goods & services — the wedge, incidence & equivalence (deck; textbook §7.5)

**Core:** a tax **drives a wedge** between the price **buyers pay** ( $p_b$ ) and the price **sellers receive** ( $p_s$ ):  $p_b = p_s + t$ . It raises the buyer price, lowers the seller price, cuts the quantity traded, and creates a **deadweight loss**. The split of the burden — **tax incidence** — depends on the **relative elasticities**, and (under perfect competition) **not** on who is legally taxed.

- **Two types of tax (deck):**
  - **Unit (specific) tax** — a fixed amount  $t$  per unit (per litre of petrol, per cigarette).

- **Ad valorem tax** ("sales tax"/VAT) — a **fraction  $\alpha$  of the price/spend** (government keeps a share of each euro spent).
- **The wedge (worked unit-tax example,  $t = €3$ )**: with  $Q_D = 23 - p_b$ ,  $Q_S = -1 + p_s/2$ , and  $p_b = p_s + 3$ , solving gives  $p_s = 14$ ,  $p_b = 17$ ,  $Q = 6$ . Buyers pay **€3 more** than sellers receive; quantity falls from the no-tax equilibrium.
- **Tax revenue** =  $t \times Q = 3 \times 6 = €18$ . The rest of the surplus lost is **deadweight loss** (here  $3 \times (7-6)/2 = €1.5$ ) — trades that *would* have happened but now don't.
- **Legal incidence  $\neq$  economic incidence (the key result)**. It **doesn't matter** whether the tax is collected from **firms** (supply shifts up by  $t$ ) or **consumers** (demand shifts down by  $t$ ) — the quantity, the prices actually paid/received, and the DWL are **identical**. *All that matters is that the tax drives a wedge.* (Perfect-competition assumptions make this exact: no transaction costs, perfect information.)



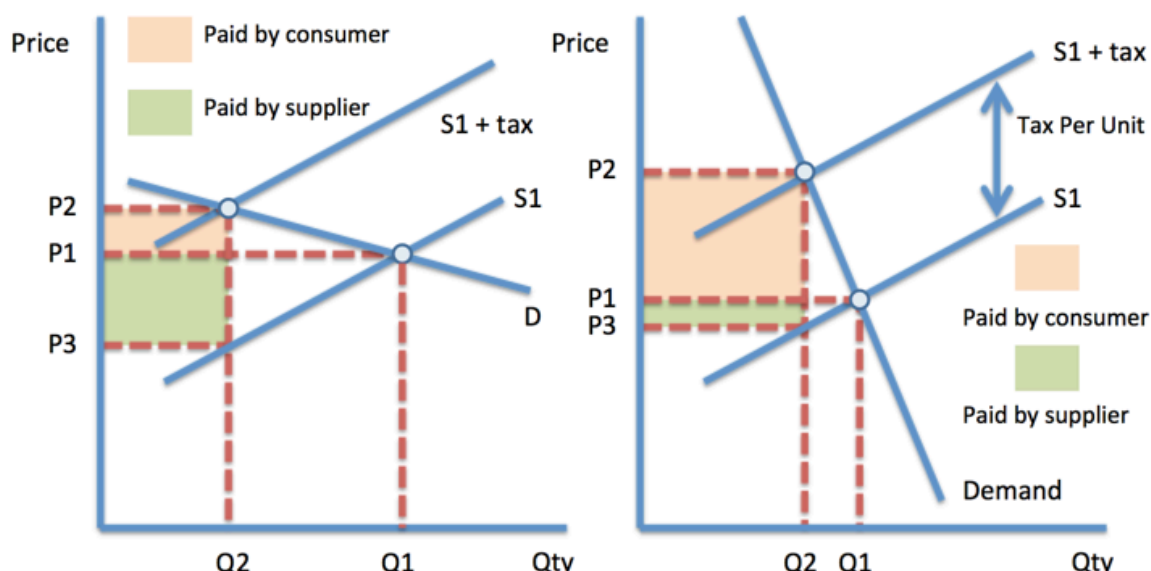
**How to read it (tax equivalence).** Both panels impose the **same \$0.55 tax** on avocados. (a) collects it from **firms** → the supply curve shifts **up** by  $t$ ; (b) collects it from **consumers** → the demand curve shifts **down** by  $t$ . In **both** cases the result is identical:  **$Q$  falls to 74, buyers pay 2.15, sellers receive 1.60**, and the wedge between them is exactly **\$0.55**. → *The economics doesn't care who writes the cheque.*

- **What *does* decide the split: relative elasticities.** The **more inelastic** side of the market bears **more** of the tax (it can't escape by changing quantity). The **more price-sensitive (elastic)** side is affected **less** on price, **more** on quantity.

## Indirect Taxes with Different Coefficient of PED

If the co-efficient of price elasticity of demand  $>1$ , then most of the burden of an indirect tax will be absorbed by the supplier

If the co-efficient of price elasticity of demand  $<1$ , most of an indirect tax can be passed on to the final consumer



**How to read it (incidence & elasticity).** The tax raises the supply curve to  $S + \text{tax}$ . The **new higher price P2** is what **consumers** pay; **P3** is what **suppliers** keep. The split of the rectangle between them is the **incidence**:

- **Left — elastic demand ( $PED > 1$ ):** consumers easily cut back, so firms **can't raise the price much** → the **supplier absorbs most** of the tax.
- **Right — inelastic demand ( $PED < 1$ ):** consumers keep buying regardless, so the firm **passes most** of the tax through → the **consumer bears most** of it.
- **Policy take-aways:** (1) "sin taxes" on inelastic goods (cigarettes, alcohol) fall mostly on **consumers** and raise a lot of revenue with **little DWL** (quantity barely moves) — which is *why* governments like them, and why they can be **regressive** (hit the poor). (2) We **prefer taxes on inelastic bases** — they distort choices least.
- **Putting a number on the split — the pass-through formula.** The diagrams say *which* side pays more; this formula says *how much*. Each side's share of the tax is the **other side's** elasticity over the total:

### TAX INCIDENCE (PASS-THROUGH)

$$\text{consumer share} = E_S \div (E_S + |E_D|) \quad \cdot \quad \text{producer share} = |E_D| \div (E_S + |E_D|)$$

- **Read it as "the other side's elasticity on top":** the **consumer** share carries  **$E_S$**  (supply's elasticity); the **producer** share carries  **$|E_D|$**  (demand's elasticity). Same denominator; the two shares **add to 1** (they split the whole tax — a handy check). - **Why the "other" side's elasticity?** A side pays a big share when it **can't escape** — i.e. when it's inelastic **and the other side is elastic**. If

supply is very elastic (big  $E_S$ ), sellers walk away rather than absorb the tax, so it lands on **consumers**  $\rightarrow E_S$  on top of the consumer share. This is the same "inelastic side bears more" rule from the diagrams, now as an equation. (Worked in [practice problem 11](#micro3-practice).) - **Subsidies are the mirror image:**  $p_s = p_b + s$  — a subsidy lowers the buyer price, raises the seller price, **expands** quantity above the efficient level, and also creates a **deadweight loss** (units produced that cost more than consumers value them). Its incidence, too, depends on elasticities (inelastic demand — e.g. emergency care, primary education —  $\rightarrow$  small DWL). - **Worked example** (subsidy  $s = €3$ ): **same market** ( $Q_D = 23 - p_b$ ,  $Q_S = -1 + p_s/2$ ) but now  $p_s = p_b + 3$   $\rightarrow$  solving gives  $p_b = 15$ ,  $p_s = 18$ ,  $Q = 8$  — buyers pay **less** and sellers get **more**, and quantity **rises** above the no-subsidy level ( $6 \rightarrow 8$ ). The subsidy costs the government  $s \times Q = 3 \times 8 = €24$ , part of which is **deadweight loss**.

## Cases & examples

Read each as a worked use of an elasticity number (or a tax result).

- **Sugary-soda tax (Allcott et al. 2019)**. Own-price elasticity  $\approx -1.37$  (elastic). A 5% price rise  $\rightarrow$   $\sim 7\%$  fall in quantity; to cut demand 15% you'd raise price  $\sim 11\%$ .  $\rightarrow$  *Illustrates: elasticity as the **design tool** for a corrective ("sin") tax.*
- **World cocoa demand**. Inelastic ( $\approx -0.19$  to  $-0.96$ ; policy estimate  $-0.34$ ).  $\rightarrow$  *Illustrates: **inelastic demand**  $\rightarrow$  supply shocks hit price, so bad harvests swing incomes for growers in Ghana / Côte d'Ivoire.*
- **US beer — brand vs category (Miller et al. 2021)**. Product-level elasticity  $\approx -4.74$ , category  $\approx -0.60$ .  $\rightarrow$  *Illustrates: **aggregation** — a single brand has close substitutes (elastic), the category doesn't (inelastic).*
- **Sweden–Denmark cross-border alcohol (Asplund et al. 2007)**. Swedes drive to Denmark for cheaper drink. Estimated for spirits: **own-price**  $-1.3$  (elastic), **cross-price (Danish price)**  $+0.3$  (substitute), **income**  $+1.4$  (normal good).  $\rightarrow$  *Illustrates: **all three elasticities at once** — own, cross, income — from real data.*
- **Electricity demand (Labandeira et al. 2017)**.  $\approx -0.24$  short-run vs  $-0.6$  long-run.  $\rightarrow$  *Illustrates: demand is **more elastic in the long run** (time to adjust equipment/habits).*

## Formula sheet — quick reference

| Quantity                   | Formula                                                             | Sign / meaning                                           |
|----------------------------|---------------------------------------------------------------------|----------------------------------------------------------|
| Price elasticity of demand | $E_D = \% \Delta Q_D \div \% \Delta p$                              | <b>negative</b> ; $ E  > 1$ elastic, $ E  < 1$ inelastic |
| Point elasticity (linear)  | $E_D = (dQ/dp) \cdot (p/Q)$                                         | varies along the curve; more elastic at higher p         |
| Constant-elastic demand    | $Q = a \cdot p^\epsilon \rightarrow \ln Q = \ln a + \epsilon \ln p$ | elasticity = $\epsilon$ everywhere                       |
| Cross-price elasticity     | $E_{\text{cross}} = \% \Delta Q_A \div \% \Delta p_B$               | <b>+ substitutes, – complements</b>                      |
| Income elasticity          | $E_{\text{income}} = \% \Delta Q \div \% \Delta I$                  | <b>+ normal, – inferior</b> ; food < 1 (Engel)           |
| Supply elasticity          | $E_S = \% \Delta Q_S \div \% \Delta p$                              | <b>positive</b>                                          |
| Tax wedge                  | $p_b = p_s + t$                                                     | buyers pay more, sellers get less; DWL                   |
| Tax revenue                | $t \times Q_{\text{taxed}}$                                         | rectangle; rest of loss = deadweight loss                |
| Subsidy wedge              | $p_s = p_b + s$                                                     | expands Q above efficient; also a DWL                    |

## Exam pointers

- Compute an elasticity from two (p, Q) points and **name** it (elastic/inelastic/unit) — the –1/3 example.
- Use the point-elasticity trick  $E_D = (dQ/dp) (p/Q)$  on a **linear demand** and show elasticity **changes along the curve** (more elastic at higher price).
- Give the **sign rules** for cross-price (substitutes/complements) and income (normal/inferior) elasticities, with an example each.
- Solve a **unit-tax market** ( $p_b = p_s + t$ ) for  $p_b$ ,  $p_s$ ,  $Q$ , compute **revenue** and **deadweight loss** — the t=€3 example.
- State and **explain** the two big tax results: **legal ≠ economic incidence** (doesn't matter who's taxed) and **incidence depends on relative elasticities** (inelastic side pays more); use the sin-tax application.
- Explain **why elasticity is unit-free** and why demand is **more elastic in the long run** and for narrower product categories.

**Bridge.** Lecture 3 turns the Chapter 2 model from *qualitative* ("the curve shifts left") into *quantitative* ("by how much, and split how between price and quantity"). Elasticity is also the hinge to later chapters: **firm pricing** (a monopolist's markup depends on  $E_D$ ), **tax policy & deadweight loss** (Ch. 7), and **who gains from trade**.

## ★ Practice problems — the whole chapter (with worked solutions)

A set covering **every calculation** in this lecture. **Try each closed-book first**, then check against the worked solution. (Exam is open-book, no AI — so the skill being tested is doing these fast and correctly, not memorising.)

What you must be able to calculate (each maps to a problem below):

| Skill                                                              | Formula                                                                                 | Problem |
|--------------------------------------------------------------------|-----------------------------------------------------------------------------------------|---------|
| Own-price elasticity from two points + name it                     | $E_D = \% \Delta Q \div \% \Delta p$                                                    | 1       |
| Predict $\% \Delta Q$ from $\% \Delta p$ (and reverse)             | $\% \Delta Q = E_D \cdot \% \Delta p$                                                   | 2       |
| Point elasticity on a linear curve (+ find the unit-elastic point) | $E_D = (dQ/dp) \cdot (p/Q)$                                                             | 3       |
| Elasticity $\rightarrow$ total revenue (raise or cut price?)       | $TR = p \cdot Q$                                                                        | 4       |
| Constant-elastic demand                                            | $Q = a \cdot p^\epsilon$                                                                | 5       |
| Cross-price elasticity + classify                                  | $E_{cross} = \% \Delta Q_A \div \% \Delta p_B$                                          | 6       |
| Income elasticity + classify                                       | $E_{income} = \% \Delta Q \div \% \Delta I$                                             | 7       |
| Supply elasticity                                                  | $E_S = \% \Delta Q_S \div \% \Delta p$                                                  | 8       |
| Unit tax: solve $p_b$ , $p_s$ , $Q$ , revenue, DWL, incidence      | $p_b = p_s + t$                                                                         | 9       |
| Subsidy: solve $p_b$ , $p_s$ , $Q$ , cost, DWL                     | $p_s = p_b + s$                                                                         | 10      |
| Incidence from elasticities (pass-through)                         | consumer share = $E_S \div (E_S +  E_D )$ · producer share = $ E_D  \div (E_S +  E_D )$ | 11      |
| Equilibrium + a price control (Ch. 2 recap)                        | set $Q_D = Q_S$                                                                         | 12      |

### Problems

1. A bakery raises bread from €4  $\rightarrow$  €5; weekly sales fall from 100  $\rightarrow$  80 loaves. Find the price elasticity of demand and say whether demand is elastic, inelastic, or unit-elastic.
2. (a) The own-price elasticity of petrol is  $-0.3$ . If the price rises 10%, by how much does quantity demanded change? (b) A cinema wants ticket sales to rise 9%; demand elasticity is  $-1.5$ . What price change is needed?

3. Demand is  $Q = 120 - 4p$ . (a) Find  $E_D$  at  $p = 10$ . (b) Find  $E_D$  at  $p = 20$ . (c) At what price is demand **unit-elastic**?
4. Using the same demand  $Q = 120 - 4p$ : at  $p = 10$  and at  $p = 20$ , should the firm **raise or cut** price to increase **total revenue**? At what price is revenue **maximised**?
5. Demand is  $Q = 500 \cdot p^{(-2)}$ . (a) What is the elasticity of demand? (b) Find quantity at  $p = 5$  and  $p = 10$ . (c) If price rises 1% from €5, roughly what is  $\% \Delta Q$ ?
6. Classify each pair as substitutes, complements, or unrelated, and give  $E_{\text{cross}}$ : (a) price of **tea** +8% → demand for **coffee** +4%; (b) price of **printers** +10% → demand for **ink** +6% **fall**; (c) price of **umbrellas** +10% → demand for **coffee** 0%.
7. Classify the good and give  $E_{\text{income}}$ : (a) income +5% → **restaurant meals** +15%; (b) income +10% → **bus tickets** -4%; (c) income +8% → **bread** +2%.
8. When the price rises from €8 → €10, quantity supplied rises from 60 → 69. Find the price elasticity of supply and classify it.
9. A market has  $Q_D = 100 - 4p$  and  $Q_S = -20 + 2p$ . (a) Find the pre-tax equilibrium. (b) A **unit tax**  $t = €6$  is imposed. Solve for  $p_b$ ,  $p_s$ , and  $Q$ . (c) Find **tax revenue** and the **deadweight loss**. (d) How is the burden split between buyers and sellers, and **why**?
10. Same market ( $Q_D = 100 - 4p$ ,  $Q_S = -20 + 2p$ ). Now the government pays a **subsidy**  $s = €6$  per unit. Solve for  $p_b$ ,  $p_s$ ,  $Q$ , the **cost to government**, and the **deadweight loss**.
11. At the pre-tax equilibrium of problem 9, the elasticities are  $E_D = -4$  and  $E_S = +2$ . Use the pass-through formula to find the **share of the tax borne by consumers vs producers**, and check it against your answer to 9(d).
12. (Ch. 2 recap.)  $Q_D = 80 - 2p$ ,  $Q_S = -10 + p$ . (a) Find equilibrium  $p^*$  and  $q^*$ . (b) The government sets a **price ceiling** at €25. Compute the resulting **shortage**.

## Solutions

1.  $\% \Delta Q = (80 - 100)/100 = -20\%$ ;  $\% \Delta p = (5 - 4)/4 = +25\%$ .

$$E_D = -20\% \div +25\% = -0.8$$

$|-0.8| < 1 \rightarrow$  **\*\*inelastic\*\*** (quantity moves less than price). \*(Note: revenue actually rises — see problem 4's logic.)\*

2. (a)  $\% \Delta Q = E_D \cdot \% \Delta p = -0.3 \times 10\% = -3\%$  (quantity falls 3%). (b) rearrange:  $\% \Delta p = \% \Delta Q \div E_D = 9\% \div (-1.5) = -6\% \rightarrow$  the cinema must **cut** price 6%.
3. Slope  $dQ/dp = -4$  (constant). Use  $E_D = (dQ/dp) \cdot (p/Q)$ .
- (a)  $p = 10 \rightarrow Q = 120 - 40 = 80 \rightarrow E_D = -4 \times 10/80 = -0.5$  (inelastic).
  - (b)  $p = 20 \rightarrow Q = 120 - 80 = 40 \rightarrow E_D = -4 \times 20/40 = -2$  (elastic).
  - (c) unit-elastic: set  $-4 \cdot (p/Q) = -1 \rightarrow Q = 4p \rightarrow 120 - 4p = 4p \rightarrow 8p = 120 \rightarrow p = 15$  ( $Q = 60$ ). (On any straight-line demand, the unit-elastic point is the **midpoint**.)

#### 4. Revenue $TR = p \cdot Q$ .

- At  $p = 10$  (inelastic,  $|E| < 1$ ): raising price **raises** revenue. Check:  $p=10 \rightarrow TR = 10 \times 80 = 800$ ;  $p=11 \rightarrow Q = 76$ ,  $TR = 836$ .  $\rightarrow$  **raise price**.
- At  $p = 20$  (elastic,  $|E| > 1$ ): cutting price **raises** revenue. Check:  $p=20 \rightarrow TR = 800$ ;  $p=19 \rightarrow Q = 44$ ,  $TR = 836$ .  $\rightarrow$  **cut price**.
- Revenue is maximised at the unit-elastic point**  $p = 15$ :  $TR = 15 \times 60 = 900$ . (Rule:  $|E| < 1 \rightarrow$  raise price;  $|E| > 1 \rightarrow$  cut price; max TR where  $|E| = 1$ .)

5. (a) For  $Q = a \cdot p^\epsilon$ , the elasticity is the exponent:  $E_D = -2$  (constant at every price). (b)  $p = 5 \rightarrow Q = 500/25 = 20$ ;  $p = 10 \rightarrow Q = 500/100 = 5$ . (c)  $\% \Delta Q \approx E_D \cdot \% \Delta p = -2 \times 1\% = -2\%$ .

6.  $E_{cross} = \% \Delta Q_A \div \% \Delta p_B$ . (a)  $+4 \div +8 = +0.5 \rightarrow$  **substitutes** (tea dearer  $\rightarrow$  more coffee). (b)  $-6 \div +10 = -0.6 \rightarrow$  **complements** (printers dearer  $\rightarrow$  less ink). (c)  $0 \div +10 = 0 \rightarrow$  **unrelated**.

7.  $E_{income} = \% \Delta Q \div \% \Delta I$ . (a)  $+15 \div +5 = +3 \rightarrow$  **normal, and a luxury** ( $E > 1$ ). (b)  $-4 \div +10 = -0.4 \rightarrow$  **inferior**. (c)  $+2 \div +8 = +0.25 \rightarrow$  **normal necessity** ( $0 < E < 1$ ; cf. Engel's law for food).

8.  $\% \Delta Q_S = (69-60)/60 = +15\%$ ;  $\% \Delta p = (10-8)/8 = +25\%$ .

$$E_S = +15\% \div +25\% = +0.6$$

$0 < E_S < 1 \rightarrow$  **inelastic supply** (output hard to expand quickly).

9. (a) **Pre-tax**: set  $Q_D = Q_S$ :  $100 - 4p = -20 + 2p \rightarrow 120 = 6p \rightarrow p^* = 20, Q^* = 20$ .

- (b) **With**  $t = 6$ , use  $p_b = p_s + 6$ ,  $Q_D = 100 - 4p_b$ ,  $Q_S = -20 + 2p_s$ . Set equal:  $100 - 4(p_s + 6) = -20 + 2p_s \rightarrow 76 - 4p_s = -20 + 2p_s \rightarrow 96 = 6p_s \rightarrow p_s = 16, p_b = 22, Q = 100 - 4(22) = 12$ . (check  $Q_S = -20 + 32 = 12 \checkmark$ )
- (c) **Tax revenue** =  $t \times Q = 6 \times 12 = \text{€}72$ . **DWL** =  $\frac{1}{2} \times t \times (Q^* - Q_{tax}) = \frac{1}{2} \times 6 \times (20 - 12) = \text{€}24$ .
- (d) Buyers bear  $p_b - p^* = 22 - 20 = \text{€}2$ ; sellers bear  $p^* - p_s = 20 - 16 = \text{€}4$  (they sum to  $t = 6$ ). **Sellers bear more because supply is the more inelastic side** (slope 2 vs demand's 4  $\rightarrow$  demand is more elastic, so it escapes more of the tax onto quantity).

10. **With**  $s = 6$ , use  $p_s = p_b + 6$ :  $100 - 4p_b = -20 + 2(p_b + 6) \rightarrow 100 - 4p_b = -8 + 2p_b \rightarrow 108 = 6p_b \rightarrow p_b = 18, p_s = 24, Q = 100 - 4(18) = 28$ . (check  $Q_S = -20 + 48 = 28 \checkmark$ ) — buyers pay less ( $18 < 20$ ), sellers get more ( $24 > 20$ ), and **Q rises** ( $20 \rightarrow 28$ ). **Cost to government** =  $s \times Q = 6 \times 28 = \text{€}168$ . **DWL** =  $\frac{1}{2} \times s \times (Q_{sub} - Q^*) = \frac{1}{2} \times 6 \times (28 - 20) = \text{€}24$  (over-production: units that cost more than they're worth).

11. Pass-through shares:

#### INCIDENCE FROM ELASTICITIES

$$\text{consumer share} = E_S \div (E_S + |E_D|) \quad \cdot \quad \text{producer share} = |E_D| \div (E_S + |E_D|)$$

With  $E_S = 2$  and  $|E_D| = 4$ : **consumer share** =  $2 \div (2+4) = 1/3$ , **producer share** =  $4 \div 6 = 2/3$ . Of the €6 tax that's **€2 on buyers, €4 on sellers** — exactly matching 9(d).  $\checkmark$  (Rule: the **more inelastic** side pays the bigger share.)

12. (a)  $80 - 2p = -10 + p \rightarrow 90 = 3p \rightarrow p^* = 30, q^* = 20$ . (b) At the ceiling  $p = 25$ :  $Q_D = 80 - 50 = 30$ ,  $Q_S = -10 + 25 = 15 \rightarrow \text{shortage} = Q_D - Q_S = 30 - 15 = 15 \text{ units}$ .

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## ★ Fun facts & memorable details (Lecture 3)

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Sticky bits from Elasticities & Taxes.

- **It doesn't matter who you tax.** In a competitive market the burden of a tax is **identical** whether the government bills the shop or the shopper — "all that matters is that the tax drives a wedge." (This is why arguing about *who* a tax is "on" often misses the point.)
  - **Why cigarette & alcohol taxes raise so much money:** demand is **inelastic**, so people keep buying — the tax is **passed to consumers** with little drop in quantity (and is often **regressive**).
  - **The soda-tax number:** the own-price elasticity of sugary drinks ( $\approx -1.37$ ) is a real input to real policy — a 5% price rise cuts consumption  $\sim 7\%$ .
  - **Engel's law (1857):** the **richer** you get, the **smaller** the share of income you spend on **food** — one of the oldest empirical regularities in economics, and still used to gauge living standards.
  - **Pharma = "defensive stock":** demand for medicine barely moves with income (income-inelastic), so drug companies hold up in recessions.
  - **Elasticity has a time dimension:** when petrol jumps, you can't do much *this week* (inelastic), but over years you buy a smaller car and move closer to work (elastic) — the "short-run vs long-run" split.
  - **Brand vs category:** demand for *a* beer is wildly elastic ( $\approx -4.7$  — switch brands!), but demand for *beer* is inelastic ( $\approx -0.6$ ) — same drink, opposite elasticity, just different zoom level.
  - **Swedes really do drive to Denmark for cheap booze** — enough that economists measured the cross-border **cross-price elasticity** of spirits ( $+0.3$ ) from the data (Asplund et al. 2007).
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